Local-First Voice AI: Build anOffline Assistant with Pipecat, Ollama, and RAG — Workshop Prerequisites

Fully-offline voice agent on Pipecat (local STT → LLM → TTS). Everything below is downloaded **once, while online**; the Voice AI Bot itself runs with networking off.

1 Downloadable prerequisites

Software

Item	Size (approx.)	Platform notes
uv (Python package manager)	~40 MB	Same one-line installer on macOS, Linux, and WSL2
Ollama (serves the local LLMs)	~0.3–1 GB	macOS app, Linux install script; inside WSL2 use the Linux script
Git + the LO-FI repository	< 50 MB	Git is preinstalled on macOS/most Linux
Python 3.13 + project dependencies	~1.3 GB	Pulled automatically by `uv sync`
PortAudio (system library; optional)	< 5 MB	macOS: `brew install portaudio` · Linux/WSL2: `apt install portaudio19-dev`

Models (one-time downloads)

Role	Model	Download	Fetch command	Platform notes
LLM (conversation + tool calling)	`nemotron-3-nano:4b` (Ollama)	2.8 GB	`uv run python scripts/prefetch_models.py` one command, one-time, while online — fetches every model for your platform	identical everywhere
RAG embeddings	`nomic-embed-text` (Ollama)	0.3 GB		identical everywhere
Speech-to-text	faster-whisper `distil-medium.en`	0.8 GB		CPU; auto-selected on Intel Mac / Linux / WSL2
Speech-to-text (Apple Silicon)	Whisper-MLX `large-v3-turbo`	1.6 GB		Apple Silicon only (MLX)
Text-to-speech	Kokoro (ONNX) + voices	0.35 GB		CPU; identical everywhere

> The prefetch pulls the Ollama models too (starting a temporary `ollama serve` if needed), so Ollama must be installed first. Everything lands in the repo's `./models/` directory.

> Feel free to download other models via `./lofi.sh configure -i` — `./lofi.sh configure` shows recommendations for your machine. In addition to `nemotron-3-nano:4b`, `qwen2.5:14b` (9.0 GB) and `llama3.2:1b` (1.3 GB) are interesting demo models.

2 Approximate total download size

≈ 7–8 GB per machine (≈ 5.9 GB models + ≈ 1.3 GB Python dependencies + installers). With the optional demo models (`qwen2.5:14b` + `llama3.2:1b`, ≈ 10.3 GB more): **≈ 17–18 GB**. Attendees should have **at least 12 GB of free disk** (25 GB with the optional models). Strongly recommend downloading *before* the workshop.

3 Non-downloadable prerequisites

- **Hardware:** >=16 GB RAM recommended (8 GB works with smaller LLM tags — the repo's `configure.sh` picks a cascade that fits). Apple Silicon Macs get GPU-accelerated STT; Intel Macs, Linux, and WSL2 run STT on CPU.
- **Operating system:** macOS or Linux. Windows is supported *only via WSL2* (all Linux instructions apply verbatim inside WSL2; keep the repo on the Linux filesystem, not `/mnt/c`). Native Windows is not supported.
- **Audio:** a working microphone, and **headphones** (optional but recommended). WSL2 users should use the browser client (not native audio).
- **Web Browser:** Chrome / Chromium-based browser recommended.
- **Skills:** comfort running commands in a terminal.